Selection on Observables III: Propensity Scores and Weighting

PS200C, Spring 2018

Chad Hazlett

UCLA

chazlett@ucla.edu

Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
4. Instrumental Variables
5. Regression Discontinuity
6. Difference in Differences and Synthetic Control
7. Sensitivity and Bounds

We have to keep saying this...

Assumption: Conditional Ignorability

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

(a.k.a. exogeneity, unconfoundedness, selection on observables, no omitted variables)

So we have "experiments" within each stratum of X, but how to estimate an average effect as if we did it strata-wise then averaged? ("conditioning on X")

- We took very literal approach with sub-classiciation and matching
- Now, let's think about re-weighting in ways that make the overall distribution of treated and control more similar
- This will allow us to estimate things as if we have done so conditionally on X then averaged over some density in X

Why Weighting?

A few ways to come up with the weighting idea...

First, we have been getting conditional treatment effect estimates, $\hat{\mathbb{E}}[Y_{1i} - Y_{0i}|X]$, then putting these together by some distribution of X.

But, *if* we could make treated and control have same distribution in X first, we could take a simple average.

Think about what it would mean to do sub-classification estimator if $p(X|D = 1) = p(X|D = 0)$ or matching if you *already have* an effectively matched sample.

Why Weighting?

More formally,

- We have conditional treatment effects in each stratum of X:

$$\mathbb{E}[\tau|X] = \mathbb{E}[Y_i|D_i = 1, X_i] - \mathbb{E}[Y_i|D_i = 0, X_i]$$

- Then we integrate them using some $p(X^\star)$:

$$\begin{aligned}\mathbb{E}_X\mathbb{E}[\tau|X] &= \int \mathbb{E}[Y_i|D_i = 1, X_i] - \mathbb{E}[Y_i|D_i = 0, X_i]p(X^\star)dx \\ &= \int \mathbb{E}[Y_i|D_i = 1, X_i]p(X^\star)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X^\star)dx \\ &= \text{mean for treated minus controls when same density, p(X*)}\end{aligned}$$

Suggests "moving" treated and control to same density in X, then taking difference in means.

What Weights?

Recalling the last, we want something like:

$$\int \mathbb{E}[Y_i|D_i = 1, X_i]p(X^\star)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X^\star)dx$$

But if we just do difference in means (w/o) conditioning, we are getting:

$$\int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i|D = 1)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i|D_i = 0)dx$$

Consider weighting by $\frac{p(D_i)}{p(D_i|X_i)}$:

$$\begin{aligned}
\mathbb{E}[\hat{\tau}_{IPW}] &= \int \mathbb{E}[Y_i|D_i = 1, X_i]\frac{p(X|D = 1)p(D_i = 1)}{p(D_i = 1|X_i)}dx \\
&\quad - \int \mathbb{E}[Y_i|D_i = 0, X_i]\frac{p(X|D_i = 0)p(D_i = 0)}{p(D_i = 0|X_i)}dx \\
&= \int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i)dx
\end{aligned}$$

What Weights?

How convenient! These are the (stabilized) Inverse Propensity Score (IPW) weights.

- these are great in theory: no matching discrepancy to worry about, $dim(X)$ doesn't seem to matter
- but requires a (fitted) model to get $p(D_i|X_i)$, so vulnerable to misspecification
- this is one view of propensity scores, $p(D_i = 1|X_i)$. More on that soon.

Weighting without a Model

Other approaches to weighting don't require a model of the treatment assignment or outcome at all!

Rather, logic comes back to balance:

- recall, when we matched we wanted to make sure we got "balance"
- we know we want to move treated and control units to have similar distributions in X to get the conditioning done

So we only need to choose weights that "get us balance":

- typically we settle for equal means in X
- though we can try to get equal means on X^2, $X_1 X_2$, etc.
- it turns out that these get you unbiased ATT estimates when Y_{0i} is linear in the things you get mean balance on

Weighting for covariate balance

To be explicit: find weights on control units s.t. the mean of X (or other functions of X) among controls matches that of treated.

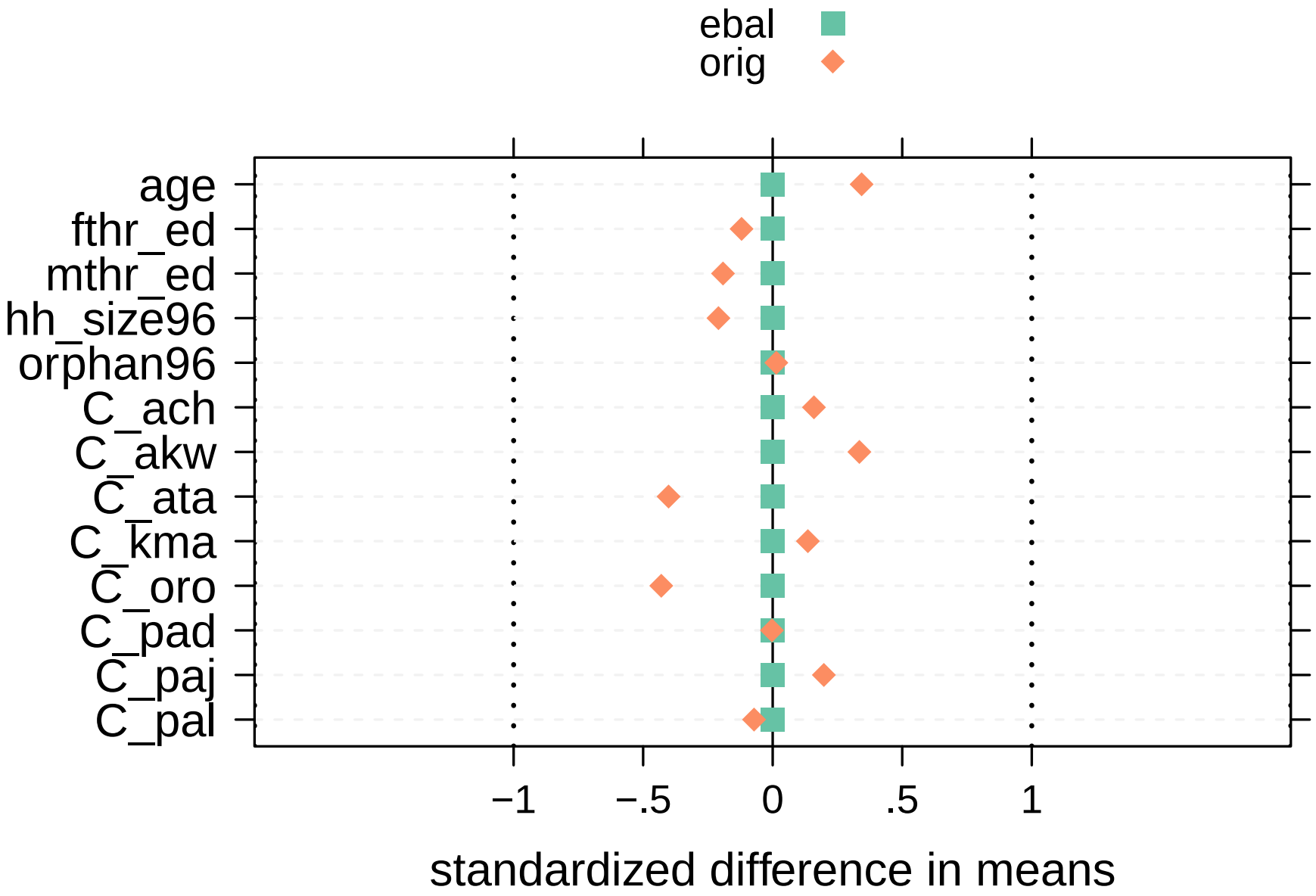
Resulting weights can be use to get weighted difference in means, or applied to any outcome model (e.g. regression)

- Could also move treated to controls (for what estimand?), or move both to overall means (for what estimand?)
- You can often find perfect mean balance, no iteration
- Standard errors are still an open question. Bootstrap likely okay.

Get equal means: entropy balancing (Hainmueller, 2011), `ebal`

To get multivariate (smoothed) balance, $\hat{p}(X|D=0) = \hat{p}(X|D=1)$, Kernel Balancing (`kbal`, see my website).

Balance on Blattman data with Ebal



Back to Propensity Scores

We just saw a kind of weight that has only to do with getting balance on X w/o contemplating a treatment assignment model.

But recall an option we saw earlier: Nature gave us

$$\int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i|D = 1)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i|D_i = 0)dx$$

Weighting by $\frac{p(D_i)}{p(D_i|X_i)}$ or equivalently $\left(\frac{p(X_i)}{p(X_i|D_i)}\right)$ gives

$$\int \mathbb{E}[Y_i|D_i = 1, X_i]\frac{p(X|D = 1)p(D_i = 1)}{p(D_i = 1|X_i)}dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]\frac{p(X|D_i = 0)p(D_i = 0)}{p(D_i = 0|X_i)}dx$$
$$= \int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i)dx$$

We call $p(D_i = 1|X_i)$ the propensity score, and $\frac{p(D_i)}{p(D_i|X_i)}$ the (stabilized) inverse propensity score weights (IPW).

Propensity Scores: Standard Motivation

A more conventional approach to the propensity score:

- May be tough to get close matches on X_i when multi-dimensional
- What if you could just match on a one-dimensional summary?

Conditioning on the Propensity Score

Under SOO and common support,

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid \pi(X_i)$$

The intuition: for each $\pi(X_i) = c$, you have a little experiment.

- Suppose you have $\pi(X_i) = c$. Among all units like this, D_i does not depend upon X_i.
- Once D_i no longer depends on X_i, it also no longer depends on $\{Y_{1i}, Y_{0i}\}$ (SOO).
- So its like random experiments within each stratum of $\pi(X_i)$.

Identification with Propensity Scores

Proof.

Show that $\Pr(D = 1 | Y_0, Y_1, \pi(X)) = \Pr(D = 1 | \pi(X)) = \pi(X)$, implying independence of (Y_0, Y_1) and D conditional on $\pi(X)$.

$$\begin{aligned}
\Pr(D = 1 | Y_1, Y_0, \pi(X)) &= \mathbb{E}[D | Y_1, Y_0, \pi(X)] \\
&= \mathbb{E}\left[\mathbb{E}[D | Y_1, Y_0, X] | Y_1, Y_0, \pi(X)\right] \text{ (LIE)} \\
&= \mathbb{E}\left[\mathbb{E}[D | X] | Y_1, Y_0, \pi(X)\right] \text{ (SOO)} \\
&= \mathbb{E}\left[\pi(X) | Y_1, Y_0, \pi(X)\right] \\
&= \pi(X)
\end{aligned}$$

Using a similar argument

$$\begin{aligned}
\Pr(D = 1 | \pi(X)) &= \mathbb{E}[D | \pi(X)] = \mathbb{E}[\mathbb{E}[D | X] | \pi(x)] \\
&= \mathbb{E}[\pi(X) | \pi(X)] = \pi(X)
\end{aligned}$$

therefore $\Pr(D = 1 | Y_1, Y_0, \pi(X)) = \Pr(D = 1 | \pi(X))$ □

Using Propensity Scores

First, estimate $P(D_i = 1|X_i) = \hat{\pi}_i$, e.g. by logit. Then condition on it:

- Matching:
  - may need a *caliper* to trim cases without common support
  - or, trim cases with extreme pscores by hand to improve common support
  - but beware, this changes your estimand
- Weighting
  - Simple IPW weights: $\frac{1}{p(D_i|X_i)} = \frac{1}{D_i\pi_i+(1-D_i)(1-\pi_i)}$
  - Stabilized IPW weights: $\frac{p(D_i)}{p(D_i|X_i)}$ aka $\frac{p(X_i)}{p(X_i|D_i)}$.
  - But if using weights, check how extreme they get and how few units are doing most of the work

Using Propensity Scores

Pros:

- Just one thing to match on
- Ignores imbalance on X's that do not predict D...very helpful for large P
- Will give you balance ("balancing property"), but *only in expectation* and if correctly estimated

Cons:

- If balancing property seen as the *goal* of pscores – by that metric it often under-performs. (See King & Nielsen 2015)
- Requires correct model and estimation of the propensity score!
- Extreme weights can cause large bias (so check)

IPW weights that also gaurantee balance?

- Covariate balancing propensity scores (CBPS; Imai, Ratkovic 2014)
- For continuous treatments, CBGPS (Fong, Hazlett, Imai 2017)

Using Propensity Scores: Blattman & Annan

(All the following posted in the markdown document on course website)

```
pi.out = glm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+C_ach+C_akw+
C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,data=dat,family="binomial"(link=logit)

matchout.pi=Match(Y=dat$educ,Tr=dat$abd,X=pi.out$fit,estimand="ATT")
summary(matchout.pi)

#Check balance
mb.out.pi =MatchBalance(match.out=matchout.pi,abd~age+fthr_ed+
mthr_ed+hh_size96+orphan96+C_ach+C_akw+C_ata+C_kma+
C_oro+C_pad+C_paj+C_pal,data=dat)

btest_after_pi=baltest.collect(mb.out.pi,var.names=varnames,after=TRUE)

balancecompare.pi=cbind(btest[,c("mean.Tr","mean.Co","T pval",
"KS pval")],btest_after_pi[,c("mean.Tr","mean.Co","T pval","KS pval")])
```

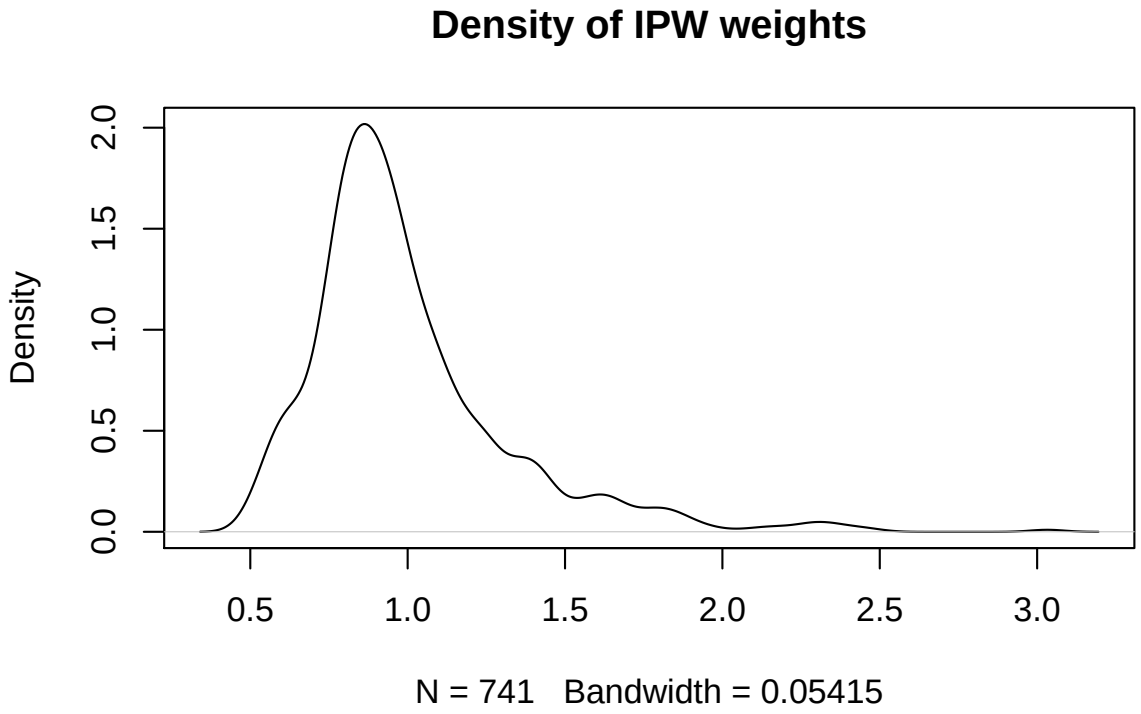
Using Propensity Scores: Blattman & Annan

```
 balancecompare.pi
          mean.Tr mean.Co T pval KS pval mean.Tr mean.Co T pval KS pval
age         21.37   20.15   0.00    0.00   21.37   21.04   0.27    0.17
fthr_ed      5.76    6.07   0.27    0.87    5.76    5.60   0.48    0.09
mthr_ed      2.09    2.49   0.09    0.35    2.09    2.40   0.11    0.20
hh_size96    8.09    8.70   0.06    0.04    8.09    8.02   0.79    0.03
orphan96     0.08    0.08   0.90      NA    0.08    0.11   0.07      NA
C_ach        0.15    0.11   0.13      NA    0.15    0.17   0.60      NA
C_akw        0.16    0.08   0.00      NA    0.16    0.17   0.62      NA
C_ata        0.10    0.20   0.00      NA    0.10    0.09   0.55      NA
C_kma        0.15    0.12   0.19      NA    0.15    0.16   0.87      NA
C_oro        0.05    0.14   0.00      NA    0.05    0.05   0.95      NA
C_pad        0.12    0.12   0.98      NA    0.12    0.09   0.10      NA
C_paj        0.15    0.10   0.06      NA    0.15    0.15   0.91      NA
C_pal        0.11    0.13   0.51      NA    0.11    0.13   0.45      NA
```

Using Propensity Scores: Blattman & Annan

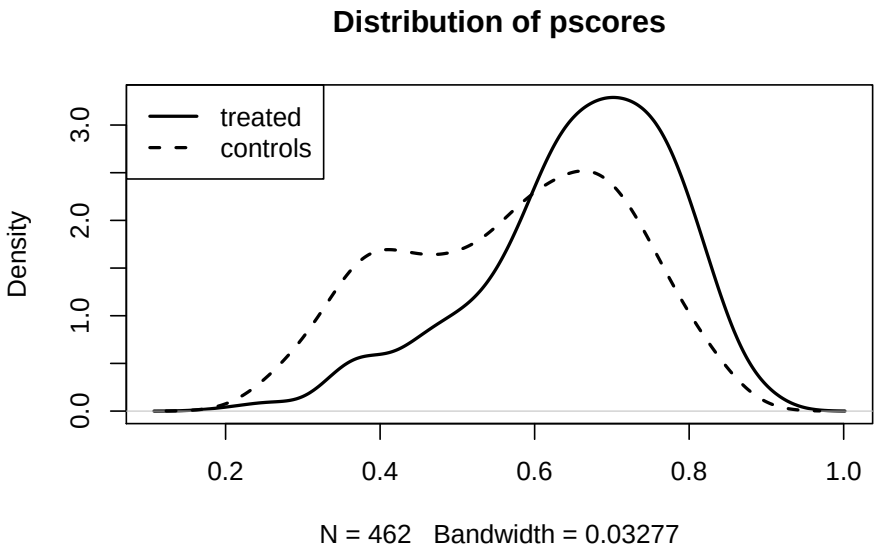
Weighting on the propensity scores (stabilized IPW):

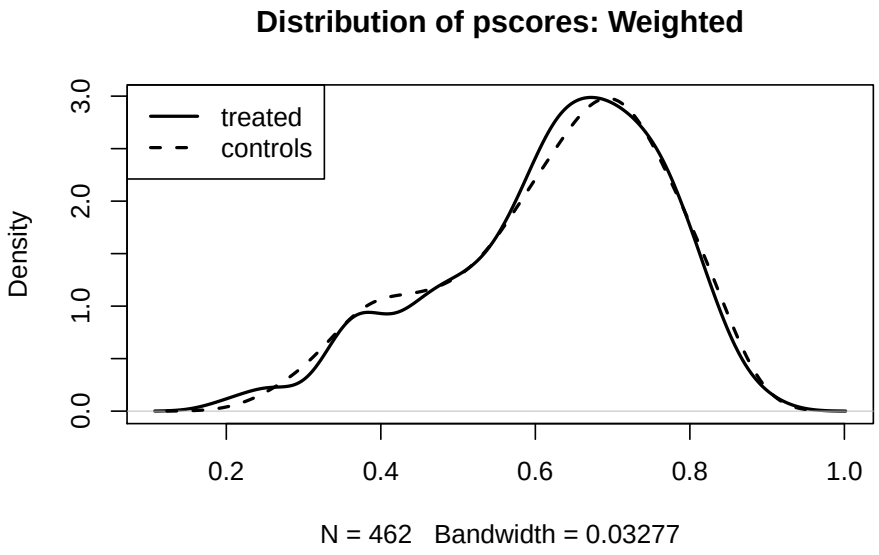
```
ps=pi.out$fit
D=dat$abd
PrD=mean(D)
IPW= (D*PrD+(1-D)*(1-PrD))/(D*ps+(1-D)*(1-ps))

#Good to check how crazy the weights are:
plot(density(IPW))
```



How does $p(\pi_i)$ look before/after weighting?

```
plot(density(pi.out$fit[D==1], weight=IPW[D==1]/sum(IPW[D==1])))
lines(density(pi.out$fit[D==0], weight=IPW[D==0]/sum(IPW[D==0])),lwd=2)
legend("topleft", legend=c("treated","controls"), lty=c(1,2))
```





Check Omnibus Balance

```
omnibus.bal = lm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+
                  C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal, data=dat)
summary(omnibus.bal)
#R2=0.08, F-statistic: 5.585 on 12 and 728 DF,  p-value: 3.303e-09


omnibus.bal.ipw = lm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+
                  C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,
                  weight=IPW, data=dat)
summary(omnibus.bal.ipw)

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.138e-01  1.058e-01   5.804 9.69e-09 ***
age          5.474e-04  3.691e-03   0.148    0.882
fthr_ed     -4.053e-04  5.352e-03  -0.076    0.940
mthr_ed     -2.116e-04  6.628e-03  -0.032    0.975
hh_size96    4.875e-04  4.689e-03   0.104    0.917
orphan96     3.979e-03  6.873e-02   0.058    0.954
---
Multiple R-squared:  0.0002747,Adjusted R-squared:  -0.0162
F-statistic: 0.01667 on 12 and 728 DF,  p-value: 1
```

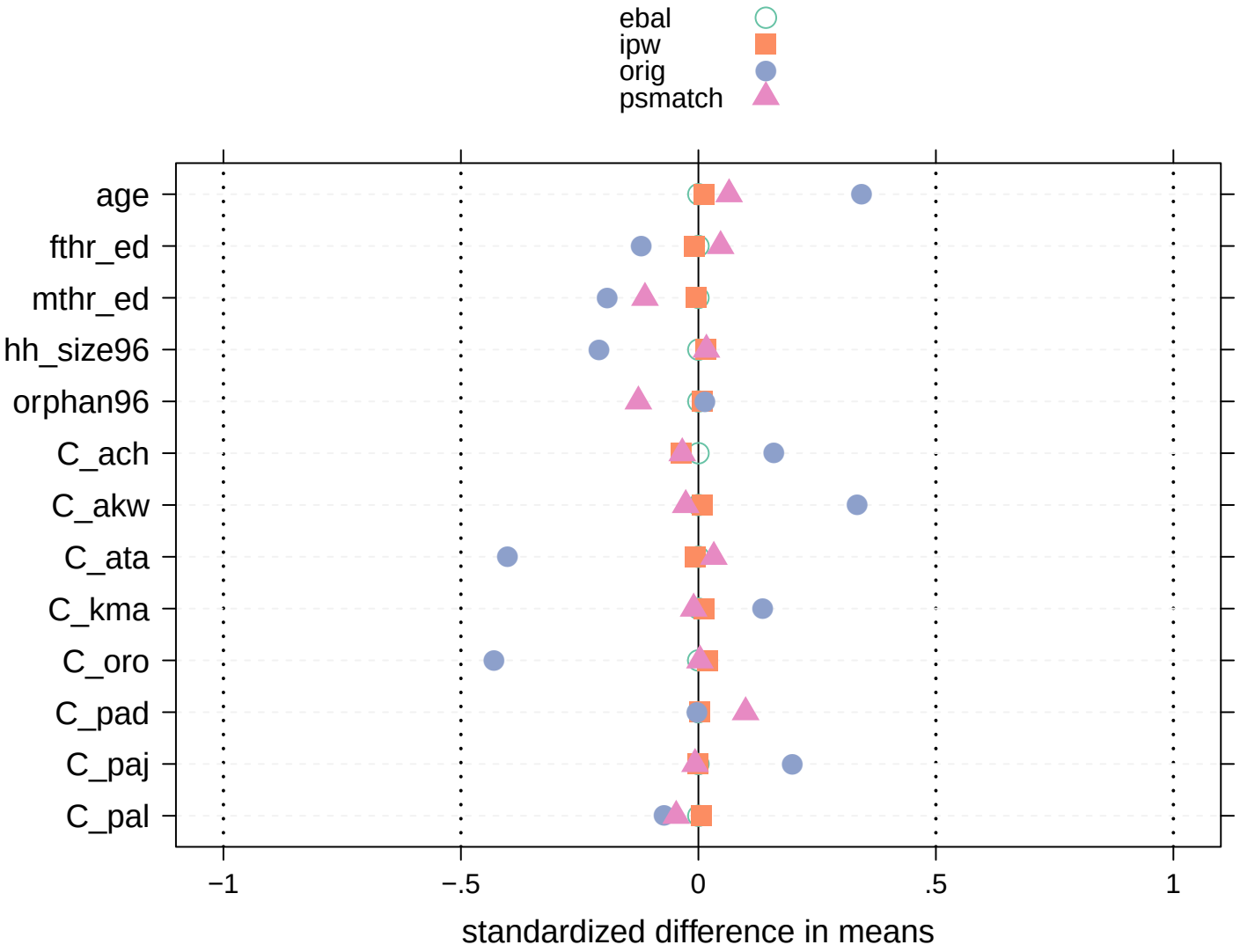
Yay!

Estimating the ATE with IPW weights

```
lm.out.naive = lm(educ~abd, data=dat)
lm.out.ipw = lm(educ~abd, weight=IPW, data=dat)
lm.out.ipw2=lm(educ~abd+age+fthr_ed+mthr_ed+hh_size96+orphan96+
    C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,weight=IPW,data=dat)
```

	Dependent variable:		
	educ		
	(1)	(2)	(3)
abd	-0.595^{***}	-0.726^{***}	-0.735^{***}
	(0.218)	(0.220)	(0.210)
Constant	7.416^{***}	7.499^{***}	6.113^{***}
	(0.172)	(0.174)	(0.610)
Observations	741	741	741
R^2	0.010	0.014	0.128
Adjusted R^2	0.009	0.013	0.112
Residual Std. Error (df = 739)	2.876	2.904	2.754
Note:		$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	

Balance Plot after IPW



Model-based Estimation of Causal Effects

Regression popular to "control for" variables. Does this allow causal inference?

Take the quantity:

$$\mathbb{E}[\tau(x)] = \mathbb{E}[Y_{1i} \mid X_i = x] - \mathbb{E}[Y_{0i} \mid X_i = x]$$

Which under CI is identifiable with estimator:

$$\mathbb{E}[\tau(x)] = \mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]$$

A model-based approach for estimating causal effects:

- Suppose $\mathbb{E}[Y_{0i}|X_i] = \mathbb{E}[Y_i \mid D_i = 0, X_i] = \beta_0 + \gamma^\top X_i$
- ...and $\mathbb{E}[Y_{1i}|X_i] = \mathbb{E}[Y_i \mid D_i = 1, X_i] = \beta_0 + \beta_1 + \gamma^\top X_i$
- Which means $\beta_1 = \mathbb{E}[\tau(X)]$
- As single model:

$$\mathbb{E}[Y_i \mid D_i, X_i] \;=\; \beta_0 + \beta_1 D_i + \gamma^\top X_i,$$

OLS as an Estimator of Causal Effects

When is $\hat{\beta}_{OLS}$ a good estimator of τ_{ATE}?

We slipped in two assumptions:

(1) Constant treatment effect. We assumed

- $\mathbb{E}[\tau(X)]$ is always one number...
- i.e., constant (expected) treatment effects (by strata of X)

(2) Linearity: Between our model and the CI assumption, we asserted

$$\mathbb{E}[Y_{id}|X_i] = \beta_0 + \beta_1 d_i + \gamma^\top X_i \quad \text{for} \quad d = 0, 1$$

Equivalently,

$$Y_{id} = \beta_0 + \beta_1 d_i + \gamma^\top X_i + \varepsilon \quad \text{for} \quad d = 0, 1$$

Noting that (2) implies (1) (such that $\beta_1 = \mathbb{E}[\tau(X)] = ATE = ...$), there are 3 possible scenarios:

1. Both (1) and (2) are true.
2. Only (1) is true.
3. Neither (1) nor (2) is true.

Case 1: Constant Effect w/ Linear Potential Outcomes

Result: If treatment effect is constant across units and potential outcomes are linear in X_i, then the OLS estimate of β_1 in the following regression model

$$Y_i = \beta_0 + \beta_1 D_i + \gamma^\top X_i + \varepsilon_i$$

is an unbiased and consistent estimator of τ_{ATE}.

Proof (just how we got here):

$$\begin{aligned}
\mathbb{E}[\beta_1] &= \mathbb{E}[Y_i|D_i = 1, X_i] - \mathbb{E}[Y_i|D_i = 0, X_i] \quad \text{(correct specification)} \\
&= \mathbb{E}[Y_{1i}|X_i] - \mathbb{E}[Y_{0i}|X_i] \quad \text{(SOO)} \\
&= \tau(X) \\
&= \tau \quad \text{(constant effect assumption)}
\end{aligned}$$

Note that if CI and linearity hold, ε cannot be related to D: traditional CIA assumption

Case 2: Constant Effect w/ Unknown Functional Form

Linearity is a very strong assumption in most practical situations.

What if $\mathbb{E}[Y_i(d)|X_i]$ is an unknown, nonlinear function of X and D but we use OLS?

Recall that OLS is still the best linear predictor in terms of MSE:

$$\hat{\beta}_{OLS} = \underset{\hat{\beta}_1}{\text{argmin}} \; \mathbb{E}\left[(Y_i - \hat{\beta}_0 - \hat{\beta}_1 D_i - \hat{\gamma}^\top X_i)^2\right]$$

Implies $\hat{\beta}_{OLS}$ is best linear approximation to the population regression function:

$$\hat{\beta}_{OLS} = \underset{\hat{\beta}_1}{\text{argmin}} \; \mathbb{E}\left[\left(\mathbb{E}[Y_i \mid D_i, X_i] - \hat{\beta}_0 - \hat{\beta}_1 D_i - \hat{\gamma}^\top X_i\right)^2\right]$$

- $\therefore$ we say "$\hat{\beta}_{OLS}$ is best linear approximation to the true treatment effect", whatever the true functional form is.
- This approximation may or may not be good! Danger of mis-specification bias.
- More flexible models (nonlinear, semi-/non-parametric, etc.) are a good idea

Case 3: Heterogeneous Treatment Effects

We typically cannot assume constant treatment effects. Is this a problem?

Recall the subclassification estimator for the ATE:

$$\hat{\tau}_{ATE} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x)$$

Similarly, the subclassification estimator for the ATT:

$$\hat{\tau}_{ATT} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x \mid D_i = 1)$$

Result: The OLS estimator can be written as a subclassification estimator, where the weights come from conditional variances of D_i in each subgroup:

$$\hat{\beta}_{OLS} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \frac{\mathrm{Var}(D_i \mid X_i = x) \Pr(X_i = x)}{\sum_{x'} \mathrm{Var}(D_i \mid X_i = x') \Pr(X_i = x')}$$

- Intuition: OLS is minimizing MSE and learns more from strata of X where π_X closer to 0.50.
- think about what this means for your estimate, especially with lack of overlap
- (See Angrist and Pishke for derivation)

OLS as a Subclassification Estimator

Estimator	Weights for Subgroups	Unbiased for
$\hat{\tau}_{ATE}$	$\Pr(X_i = x)$	τ_{ATE}
$\hat{\tau}_{ATT}$	$\Pr(X_i = x \mid D_i = 1)$	τ_{ATT}
$\hat{\beta}_{OLS}$	$\frac{\mathrm{Var}(D_i \mid X_i = x)\Pr(X_i = x)}{\sum_{x'} \mathrm{Var}(D_i \mid X_i = x')\Pr(X_i = x')}$	"$\tau_{\text{CVW-ATE}}$"

- With heterogeneous treatment effects, OLS provides an unbiased estimator for a *conditional-variance-weighted average treatment effect*.
- Estimand has a causal meaning, but difficult to interpret. Not the ATE or ATT.
- Note: $\mathrm{Var}(D_i \mid X_i = x) = \pi(x)(1 - \pi(x))$. Thus,
  - what happens when π_i is 0 or 1?
  - largest weights for strata π_i close to 0.5,
- This result assumes discrete Xs, but intuition holds for continuous Xs.
- Another option: use regression to impute missing potential outcomes. E.g. Estimate $\mathbb{E}[Y_{0i}|X_i]$ and use as counterfactuals for observed Y_{1i}. Can weight each pairwise difference as you like.

Combining Regression, Matching and Weighting

The different logics of matching, weighting, and regression can be combined to improve finite-sample performance or robustness:

- Bias-corrected matching (Abadie and Imbens 2005):
  - estimate bias due to unequal Y_{0i} of matched units due to imperfect match on X's.
  - subtract it off from the matching estimate for correction
- "Doubly-robust" estimation
  - Use weighting (e.g. IPW or balancing weights) in a regression that also includes covariates
- Matching as nonparametric preprocessing (Ho et al. 2007):
  - model-based estimation of causal effect is most likely to go wrong when it involves extrapolation due to poor overlap in covariates
  - use matching to make treatment and control groups similar
  - then run regression models to estimate causal effects

Summary: Estimation under Conditional Ignorability

- Matching, weighting and regression are main methods to estimate average causal effects when one can assume conditional ignorability
- These are estimation strategies; the validity of the identification strategy (SOO) remains a first-order concern
- No single method is dominant
- Key considerations
  - does it get you good balance?
  - is there risk of extrapolation due to non-overlap?
  - is there risk you are not doing the conditioning you mean to do? Ask yourself:
  - "what assumptions of the estimation procedure might be invalid?" (e.g. close matches, common support/overlap linearity, constant treatment effects...)